

Literature Status: Superreflexive Spaces with Δ -Points

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Status

This is a lightweight literature-status packet. It records that Question 6.1 from Abrahamsen–Lima–Martiny–Perreau, *Asymptotic geometry and delta-points*, arXiv:2203.14528, is answered by the later paper Abrahamsen–Aliaga–Lima–Martiny–Perreau–Prochazka–Veeorg, *Delta-points and their implications for the geometry of Banach spaces*, arXiv:2303.00511.

Source Question

The source paper asks:

Do all superreflexive Banach spaces fail to contain Δ -points?

In the parsed source this is Question `quest:delta_SR`, at `source.tex:2164–2169`. The paper also reformulates the issue via almost Δ -points and asks in particular about renormings of ℓ_2 at `source.tex:2450–2464`.

Supporting Answer

The supporting paper explicitly says in its introduction that Corollary 6.10 of the source paper showed finite-dimensional spaces have no Δ -points and that Question 6.1 asked whether the same holds for superreflexive spaces. It then states that Section 3 answers this question negatively.

Section 3 defines an equivalent norm on ℓ_2 and lets

$$Y = (\ell_2, \|\cdot\|).$$

Theorem 3.1 states that both $e_1 \in S_Y$ and $e_1^* \in S_{Y^*}$ are super Δ -points, and neither is a Daugavet point.

Since Y is an equivalent renorming of ℓ_2 , it is superreflexive. A super Δ -point is in particular a Δ -point. Therefore the answer to the source question is negative.

Scope

This packet does not claim a new proof. It also does not settle the Daugavet-point variant: the supporting theorem explicitly says the constructed points are not Daugavet points, and the supporting paper notes that the existence of a reflexive or superreflexive space with a Daugavet point remains open.

References

- T. A. Abrahamsen, V. Lima, A. Martiny, and Y. Perreau, *Asymptotic geometry and delta-points*, arXiv:2203.14528; Banach J. Math. Anal. 16 (2022), Paper No. 57.
- T. A. Abrahamsen, R. J. Aliaga, V. Lima, A. Martiny, Y. Perreau, A. Prochazka, and T. Veeorg, *Delta-points and their implications for the geometry of Banach spaces*, arXiv:2303.00511.